

6.5

25. $C = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$ $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$

$C^T = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}$ $\lambda = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$

$CC^T = A = \begin{bmatrix} 9 & 3 \\ 3 & 5 \end{bmatrix} \Rightarrow \lambda^T = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$
 $\Rightarrow \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix} = D$

$\text{Chol}(A) = \sqrt{D} L^T = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 0 & 3 \end{bmatrix}$

27. Using $a=2, b=4, c=10$:

$2x^2 + 8xy + 10y^2 = 2(x+2y)^2 + 6y^2$
 $= 2(x^2 + 4xy + 4y^2) + 6y^2$

$= 2x^2 + 8xy + 10y^2$

-2 Lol. forgot part of the question
 32. a. Pivots' products remain positive (pos x pos) & inverse has positive pivots. However, no symmetry.

b. Product/inverse keeps length at 1 and other combinations = 0. Comp.

c. Product and inverse remain fixed in exponential notation.

d. Determinant not guaranteed to stay above zero.

e. Inverse and Cramer's Rule remain the determinant at 1.

31.

34. $S = \begin{bmatrix} 1/2 & \sqrt{3}/2 & 1 & \sqrt{3}/2 & 1/2 \\ \sqrt{3}/2 & \sqrt{3}/2 & 0 & -\sqrt{3}/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 0 & -1 & 0 & -\sqrt{3}/2 \\ 1/2 & -\sqrt{3}/2 & \sqrt{3}/2 & -\sqrt{3}/2 & 1/2 \\ 1/2 & -\sqrt{3}/2 & 1 & -\sqrt{3}/2 & 1/2 \end{bmatrix}$

$\lambda_{1,2} = 2 \pm \sqrt{3}, \lambda_3 = 1, \lambda_4 = 2, \lambda_5 = 3$

-6 Weak proof

35. A is linearly independent.

C is positive-definite.

$A^T C A$ is well.

Since C is positive-definite

$\Rightarrow A^T C$ might change the pivots/ λ -values, but multiplying by A once more undoes these changes and det > 0 (since

linear independence)

Sol used a z-transformation

✓ I think the Rst sol has a typo
 26. $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{bmatrix} \Rightarrow L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix} \Rightarrow D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix}$

$C = \sqrt{D} L^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix}$

$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow C = \sqrt{D} L^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

29. $F_1 = \frac{x^4}{4} + x^2 + x^2 y + y^2$ $F_2 = x^3 + x y - x$

$F_x = x^3 + 2x + 2xy$ $F_x = 3x^2 + y - 1$

$F_{xx} = 3x^2 + 2 + 2y$ $F_{xx} = 6x$

$F_{xy} = F_{yx} = 2x$ $F_{xy} = F_{yx} = 1$

$F_{yy} = 2$ $F_y = 2x + 2y$ $F_y = x$ $F_{yy} = 0$

$H_1 = \begin{bmatrix} 3x^2 + 2 + 2y & 2x \\ 2x & 2 \end{bmatrix}$ $H_2 = \begin{bmatrix} 6x & 1 \\ 1 & 0 \end{bmatrix}$

$2x + 2y = 0 \Rightarrow x = -y$

$\Rightarrow x^3 + 2x + 2x^2 = 0$

$= x(x^2 + 2x + 2)$

$\Rightarrow (0, 0)$

$x = 0 \Rightarrow 0 = y - 1 \Rightarrow y = 1$

$\Rightarrow (0, 1)$

Even it's probably a saddle point. Just look at the Hessian of the problem.

33. $ABx = \lambda x$ $Bx = \lambda x$

$(ABx)^T Bx = x^T B^T A Bx$

$= x^T B^T A Bx = x^T Bx$

Energy test concludes $x^T Bx > 0$

Let $x^T Bx = R \Rightarrow x^T Bx = R \lambda$

where both $R > 0$ & $\lambda > 0$.

3.1

7. $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$C = \begin{bmatrix} c_1 & c_2 & c_3 & c_4 \end{bmatrix}$

$K = \begin{bmatrix} c_1 c_2 & -c_2 & 0 & 0 \\ c_2 c_3 & -c_3 & 0 & 0 \\ 0 & -c_3 & c_3 c_4 & -c_4 \\ 0 & 0 & -c_4 & c_4 \end{bmatrix}$

$K^{-1} = \frac{1}{5} \begin{bmatrix} 4 & 3 & 2 & 1 \\ 3 & 6 & 4 & 2 \\ 2 & 4 & 6 & 3 \\ 1 & 2 & 3 & 4 \end{bmatrix}$

with $(1, 1, 1, 1)$

$= (2, 3, 3, 2)$

10. [SEE CODE]

* I don't know why Matlab problem is one of the most flexible/creative questions!

11. [SEE CODE]

3. Forget.

$\Rightarrow [1 \ 1 \ 1] \begin{bmatrix} c_2 & c_3 & 0 \\ -c_2 & c_3 c_4 & -c_3 \\ -c_3 & c_4 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$
 $= [1 \ 1 \ 1] \begin{bmatrix} c_2 \\ c_3 \\ c_4 \end{bmatrix}$
 $\Rightarrow (c_2^{-1}, 0, c_3^{-1})$